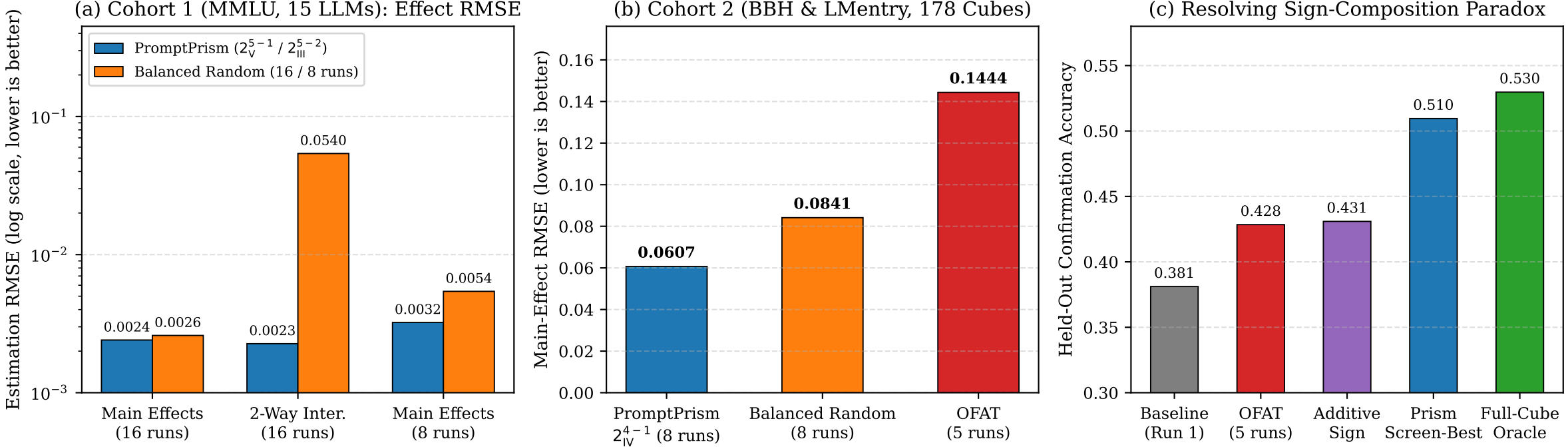
(a) Cohort 1 (MMLU, 15 LLMs): Effect RMSE
PromptPrism (2_V^{5-1} / 2_{III}^{5-2})
Balanced Random (16 / 8 runs)
Estimation RMSE (log scale, lower is better)
10^{-1}
10^{-2}
10^{-3}
0.0024
0.0026
0.0023
0.0540
0.0032
0.0054
Main Effects (16 runs)
2-Way Inter. (16 runs)
Main Effects (8 runs)
(b) Cohort 2 (BBH & LMentry, 178 Cubes)
Main-Effect RMSE (lower is better)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
0.0607
0.0841
0.1444
PromptPrism 2_{IV}^{4-1} (8 runs)
Balanced Random (8 runs)
OFAT (5 runs)
(c) Resolving Sign-Composition Paradox
Held-Out Confirmation Accuracy
0.30
0.35
0.40
0.45
0.50
0.55
0.381
0.428
0.431
0.510
0.530
Baseline (Run 1)
OFAT (5 runs)
Additive Sign
Prism Screen-Best
Full-Cube Oracle